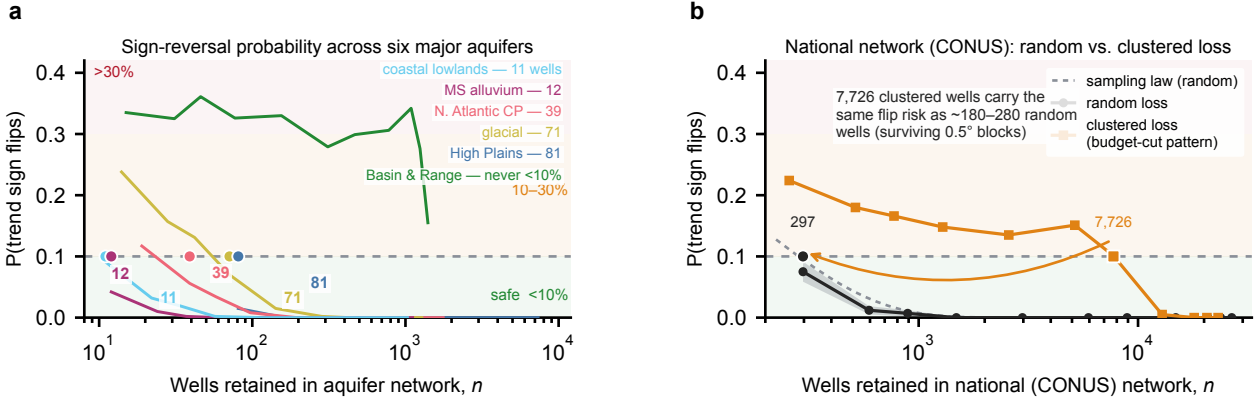
a
Sign-reversal probability across six major aquifers
>30%
coastal lowlands — 11 wells
MS alluvium — 12
N. Atlantic CP — 39
glacial — 71
High Plains — 81
Basin & Range — never <10%
10–30%
safe <10%
12
11
39
71
81
P(trend sign flips)
Wells retained in aquifer network, n
b
National network (CONUS): random vs. clustered loss
7,726 clustered wells carry the same flip risk as ~180–280 random wells (surviving 0.5° blocks)
sampling law (random)
random loss
clustered loss (budget-cut pattern)
297
7,726
P(trend sign flips)
Wells retained in national (CONUS) network, n